Modeling Smartphone Energy Consumption Based on User Behavior Data

Weizhu Qian
Univ. Bourgogne Franche Comte
UTBM - LE2I EA 7508
Belfort, France
weizhu.qian@utbm.fr

Franck Gechter
Univ. Bourgogne Franche Comte
UTBM - LE2I EA 7508
Belfort, France
franck.gechter@utbm.fr

***Abstract*—Energy consumption and power management are key issues problem when using smartphone handsets or any other connected devices. This is linked to two opposite users' expectations: having a better autonomy on one side and having devices with more computer power and bigger screens on the other side. In literature, many previous works focused on modeling the smartphone power drain at the hardware or software levels. As opposed to these approaches, we present, in this paper the construction of a handset energy model at user level based on coarse grain collected data. To that way, we adopt a crowd-sourcing approach, based on data collected by the Device Analyzer project. This data is covering all the activities/events of the user relatively to his/her smartphone. However, the data is collected within a fixed time period which involved imprecisions in the time stamp associated to each event. So as to model the energy drain and to improve events time location, we propose a top-down method based on reactive multi-agent systems paradigm. In this approach, each event is represented by an agent in a virtual world which can move along the time axis according to the behavior of its mates and which influences the battery discharging curve. As soon as the multi-agent system is stabilized, the behavioral parameters of the agents are used to build up the power model. The value of the multi-agent approach is tied to its ability to incorporate implicit information about battery drain from operating system events such as network usage. Finally, the proposed modeling method is tested on real users data and several coordination strategies are compared.**

***Index Terms*—Smartphone, energy model, multi-agent model, crowd-sourcing, user behavior**

I. Introduction

Connected objects and Smartphone are more and more widely used because of their portability and their associated various functions. As a portable device, the battery level is the first rank factor that affects the smartphone usage. Researchers are more and more interested in how the devices are behaving in the real world with various operating conditions.

In literature, most of the research papers are dealing with the power modeling issue at hardware and/or operating system level. These solutions are the most precise way to estimate the power drain. It can be done by measuring the current directly at the connection between electronic components and the battery of the handset as in [1] or by measuring the current at each component level as in [2]. By the same, some research work are focusing on monitoring the usage at operating system [3] or software levels [4]. This methods suffer from several problems. First, they required high skill technical knowledge to perform the required modifications on the devices at hardware and/or software level(s). Second, they are mainly devoted to laboratory experiments.

Few wide range experiments have been carried out so as to study how the consumer product is used in real life on a energy drain point of view and what are the behavior that have the most influence on the battery level. However, projects such as Device Analyzer, which is aimed at collecting usage data from Android devices, are important steps to help to a better understanding of this. From now on, Device Analyzer collected data from more than 31 000 users worldwide [5]. Even if a lot of participants are using Device Analyzer during a short time period, for some participants more than two years of continuous usage data are available. The data provided by device analyzer contents operating system events, stamp with their occurrence time relatively to the device time reference, and polled events such as network usage counter. These last are collected every 5 minutes so as to reduce the impact of Device Analyzer on the phenomenon one want to measure. This mixed set of events (with both real time stamps and polled time stamps) is the main difficulty in the analyze of the furnished data since it make impossible the use of classical mathematical/signal processing estimators, such as Kalman filters or Stochastic approaches, which rely generally on a regular time reference. This means that finding a solution to accurately interpolate between readings can be a valuable progress in the exploitation of the data sets.

This paper is focused on the estimation of the power consumption of a device based on polled data provided by Device Analyzer. This estimate is the first step in a better understanding of the energy consumption of smartphones tied to our daily habits. Our method is based on the processing of the user behavior data related to energy consumption provided by Device Analyzer. From this data, agents are created so as to simulate the energy drain process. Each events is associated to an agent in a virtual world that influences the natural battery level curve depending on its type and its embedded data. Then

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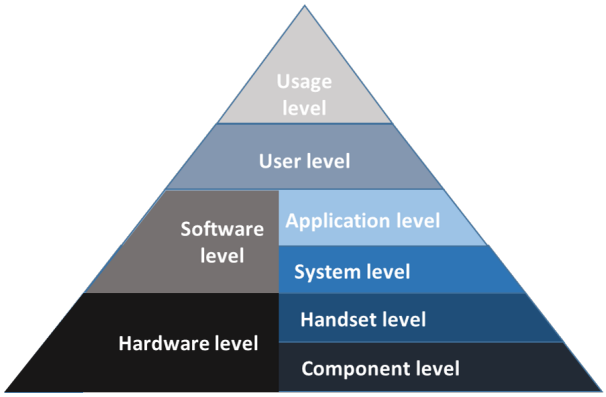

Fig. 1. Different Levels For Modeling Smartphone Energy Consumption

event-agents are interacting with their mates so as to determine its influence on battery consumption (and by extension the influence of all event of the same type) and its most probable time location.

The paper is organized as follow. Section II introduces the crowd-sourcing technique that we use for building our model and present related works on power modeling. Sections IV and V present the multi-agent system based energy consumption model and its solving process. Section VI demonstrates the smartphone energy modeling results. Finally, we give the conclusions and future work in Section VII.

II. Related works

The power modeling approaches can be sorted out into four main levels as shown in Figure 1. These levels of description are, from bottom to top, (1) the hardware level, (2) the software level, (3) the user level and (4) the usage level.

The hardware level is at the bottom of all the levels. It consists of two different levels, the component level and the handset level. At the component level, researchers break down the smartphone into hardware subcomponents, e.g., CPU, GPU, display, cellular, audio, WiFi module and bluetooth module [6], [2]. In [2], the authors use external equipment to measure the energy drain of each subcomponent as modeling benchmarks. Then they use the relationship between the utilization rate of the components and the energy cost to construct a linear regression (LR) energy model. Naturally, at the hardware level, the total energy cost of the handset is considered as the sum of each components energy cost as well as the interaction among them. However, these work are generally neglecting the interaction between components which are also influencing the global estimation.

The software level, consists of two different levels as well, the operating system level and the application level. At the system level, a frequently used method is to trace the system calls and make their relevant power states as benchmarks. A finite state machine (FSM) model is then used in some cases such as in [7]. As for the application level, the researchers' concerns are tied to the estimation of the energy exactly drained by an application, e.g. detecting the abnormal energy cost: energy bugs and energy hogs [8].

As for the user level, researchers focus on how the users' behaviors, e.g., making phone call, sending SMS, charging batteries, playing games, listening to music and taking photos, influence the energy drain of their handset devices [9], [10].

The usage level is the top level of smartphone energy modeling. At this level, not only the interaction between devices and users but the interaction with the handset and its operating the environment are taken into account. For instance, the results presented in [11] demonstrate that the strength of WiFi signal and of the 3G/4G signal can affects the handset power cost.

From this state of the art, it is clear that the power modeling of a smartphone can be considered as a tough task especially due to the complex nature of the studied system. Moreover, if the hardware level approaches seem to be relevant for a first estimation, the entanglement between different parameters at different level of abstraction lead to a necessity to design an integrative power modeling approach based on event collected from smartphone usage in its environment. For this reason, we decided to base our work on data set collected by a crowd-sourcing method and based on Device Analyzer application.

III. Crowd-sourcing of users event

[2] suggested that the energy consumed by same types of smartphone devices could differ because of the profile of their users. Thus it seems quite impossible to devise a universal energy consumption model for all handsets. [12] claims that the smartphone energy should be build in an ad-hoc way for each user. Therefore, for these reasons, we aim at modeling the smartphone energy consumption at the user level taking into account both event triggered by user and interactions between the handset and its operating environment. In this context, a crowd-sourcing technique is the adequate solution to obtain a sufficient amount of data. This is exactly what Device Analyzer is proposing.

It collects users' data such as turn on/off, charging time, incoming/outgoing calls, messages send/received, Apps usage, etc [13]. In addition, to each event is associated data which bring some more additional information. For instance a message send event is associated to the number of characters used in the text message.

As shown in table(I), the collected data of Device Analyzer can be classified into two categories: the immediate events and the polled events. The immediate events capture the user behavior at the very time where it occurs, which consists of turning on/off display, charging device, turning on/off device, etc. While the polled events record the data aggregating during a polling interval, e.g., send data bytes, received data bytes, App foreground, App service and battery level.

To avoid Device Analyzer having its own energy consumption affecting too much the measurements, therefore, for the trade-off between the the overhead of Device Analyzer and its ability of collecting data, the polling interval is set to be 5 minutes. For instance, the send data bytes is the total sending

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TABLE I
COLLECTED EVENTS

Immediate event	Polled event
Device on/off	Data received
Plugged in/out	Data send
Screen on/off	App foreground
Making call	App service
Taking photo	Brightness
	Battery level

data bytes during a polling interval of 5 minutes and the battery level is sampled only every 5 minutes. Thus, the granularity of battery level data is very coarse and the energy consumption that occurs in the polling time interval is ambiguous because of limited information of the collected data. To solve this problem, our goal is to use the energy-related data that is collected to record user behavior to construct a fine-granularity model to profile the smartphone energy cost more accurately.

IV. POWER MODEL AND MULTI-AGENT SYSTEMS

A. General overview

As explained in [14], if we break down a smartphone device into a series of subsystems at a certain level, thus the power model of a handset can be described as follow:

$$P = \sum_i (\beta_i \cdot x_i) + P_{base} + P_\epsilon \tag{1}$$

Where P is the power of the handset, β_i is the coefficient related to the subsystem energy consumption, x_i is utilization of the component, P_{base} is basic power consumption of the handset, and P_ϵ is the power noise. If we model the smartphone energy drain at the user level, the Energy consumption is the integration of power during a time period:

$$E = \sum_i \sum_j (\beta_i \cdot x_i^j) \cdot d_i + (P_{base} + P_\epsilon) \cdot D \tag{2}$$

Where E is the energy consumption of the handset, x_i^j is a scalar related to the event j and the event j belongs to category i, d_i is influence duration of the event, D is the total experiment time.

The goal of the work presented in this paper is then to find what is the influence of each event (and by extension what is the influence of each family of events) on the battery level according to its nature and its associated data. The main problem is tied to the fact that the collected data is a mix between events with a precise time stamp (relatively to device system time) and polled events. Thus, none of the classical methods used for estimating specific curves or for interpolating trends of one phenomenon cannot be adapted to our issue because they are all relying on a fixed sample time or a constant time interval. For instance, one of the condition for using a Kalman filter is the ergodicity of the process. By the same, Stochastic Approaches such as Markov models are based on a fixed time reference for the computation of transitions and observation matrices.

The key idea of our approach is to build up a virtual world, called environment, where each event collected by Device Analyzer is represented by an agent (called event agent). In this environment the battery curve is a structure made of charged particles, the trajectory of which can be influenced by a force generated by each event agent.

The power modeling is made by two steps. In the first step, agents are "trained" on several data sets so as to determine their parameters using a self organizing process. In this step, each created event-agent behavior consists in several items: (1) each agent must find its influence on the battery curve by modifying its associated parameters and measuring the generated error (2) each agent must tune its influence according to the results obtained by its mates. In the second step, we use the parameters determined in the first step to measure the error of the power model on other data sets from the same couple user/device. Section IV-B presents in details the multi-agent model.

B. A Physics inspired multi-agent model

In our multi-agent system smartphone energy model, several types of agents that have specific functions are designed to interact with each other with their environment: the event agents EA and the particle agents PA.

1) Event agent: After sorting out the events that are causing power drain from the dataset, an event agent (EA) is generated and associated to each energy-related event. Each collected event is classified into different categories according to its own representative meaning, e.g., the data sending agents refer to data sending events, the data received data agents refer to data received events and the screen on/off agents refer to screen on/off events. Once an event is modeled by an event agent, the influence of each of them can be defined.

To that way, each event agent is associated to a smart force which influence, in the agent world, is related to its influence on energy drain. A smart-force is associated to one EA that has three properties: a range R_{EA}, an offset O_{EA} and a strength S_{EA}. The range represents the time window where the agent influences the energy drain. The influence range for a immediate event is easy to be determined since it is collected by Device Analyzer. If we define the state of an immediate event such as 1,0, e.g, for the screen on/off event, when the screen is turned on, the state is 1, when it is turned off, the state is 0. Obviously, the time period where the immediate event's state switch 1 to 0 is the influence range of this event. As for the influence range of a polled event, it can be determined by (3):

$$R_{EA_k^i} = \alpha \cdot R_{EA_k} \tag{3}$$

where the event EA_k^i is a member of the event category EA_k and α is a factor which corresponds to the quantity value of the event EA_k^i, for instance, for the data send event, it is corresponding to the number of bytes sent.

The strength represents the scalar value of the smart-forces' influence on energy drain. The strength can be described as (4):

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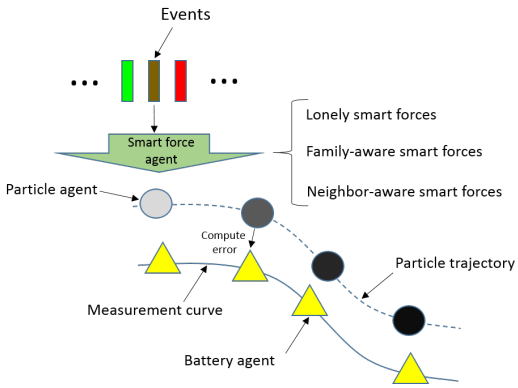

Fig. 2. MAS modeling process

$$S_{EA_k^i} = \beta \cdot S_{EA_k} \tag{4}$$

where β is the coefficient and $S_{EA_k^i}$ is the strength for a specific event event category.

The offset is used for placing the event agent on a certain location on the time line. Assuming that an event agent is collected at the time spot t_{EA}, the time window where an event agent leads to energy drain will be $[t_{EA}+O_{EA}-R_{EA}, t_{EA}+O_{EA}+R_{EA}]$.

2) Particle agent: The particle agent represents the state of discharge (SoD) of the smartphone battery. In our model, we use a beam made of a series of particles to simulate the energy drain curve. Each particle agent has three properties, position (p_x, p_y), velocity (v_x, v_y) and acceleration (a_x, a_y), to describe its movement. The particle's movement is influenced by smart-forces generated by event agents and is determined according to Newton's second law of motion.

Then, the battery level events collected by Device Analyzer are used in the training step as reference points to trigger the self organized tuning process. In the testing step, they are used for measuring the error of the model.

V. Modeling Process

A. Global overview

Figure 2 depicts the MAS model training process. User operations are recorded as event agents and associated smart-forces to influence particle agents' trajectory. Each particle agent perceives the nearest battery agent and computes the error. Errors are used as feedbacks to tune the smart-force agents' coefficients: ranges, strengths and offsets. Finally, the particle beam's trajectory model smart-phone battery level curve.

B. Autonomous tweaking strategies

To fit the battery curve accurately, the properties of the event agents need to be modified as the self-organized modeling process is carrying on. It means that their ranges will expand or shrink, their strengths will increase or decrease and their offsets will move to the left or to the right of the time line according to the feedback errors. In order to determine how to tune the parameters exactly during the model training process, we devised three different types of interacting strategies for the smart forces parameters tweaking: the lonely smart-force strategy, the family-aware smart-force strategy and the neighbor-aware smart-force strategy.

1) Lonely smart-force strategy: The lonely smart-force strategy means that there is no interaction among even agents, that is to say, each smart forces associated to an event agent EA^i has its own range R_{EA^i} and strength S_{EA^i} independently to the other. Thus, during the model training process, each smart force influences the particle agent's movement independently and modifies only its own coefficients.

2) Family-aware smart-force strategy: If the collected events are the same type of events, the events can be considered to belong to a same agent family F_k. We assume that events EA_{C_k} in the same family have the same effect on smart phone energy drain, therefore we compute the average strength S_{F_k} and the average range R_{C_k} of each agent family when tuning the smart forces coefficients.

$$S_{F_k} = \frac{1}{M}\sum_{i}^{M} S_{EA^i_{F_k}} \tag{5}$$

Where M is the number of event agent of the family F_k and $S_{EA^i_{F_k}}$ is the strength of EA^i which belongs to the family F_k.

$$R_{F_k} = \frac{1}{M}\sum_{i}^{M} R_{EA^i_{F_k}} \tag{6}$$

Where $S_{EA^i_{F_k}}$ is the strength of EA^i which belongs to the family F_k.

In this strategy, we use the average strength S_{F_k} and the average range R_{F_k} instead of each event individual range and strength for the training process.

3) Neighbor-aware smart-force strategy: The events that occur at a near zone N_k of the time line can be considered as "neighbors". Thus, the neighbor agents have the same range and strength which are the average range and average strength of the events that are in the same data zone.

$$S_{N_k} = \frac{1}{M}\sum_{i}^{M} S_{EA^i_{N_k}} \tag{7}$$

Where M is the number of event agent of the data zone N_k and $S_{EA^i_{F_k}}$ is the strength of EA^i which belongs to the family F_k.

$$R_{N_k} = \frac{1}{M}\sum_{i}^{M} R_{EA^i_{N_k}} \tag{8}$$

Where $S_{EA^i_{F_k}}$ is the strength of EA^i which belongs to the family N_k.

As in the family-aware smart-force strategy, we use the average strength S_{N_k} and average range R_{N_k} to tune the parameters of the event of the neighborhood.

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TABLE II
MODELING ERROR

Strategy	Family-aware	Lonely	Neighbor-aware
Total error	26.07%	21.03%	14.77%
Average error	1.00%	0.81%	0.57%

TABLE III
AVERAGE STRENGTH COEFFICIENTS

Events	Family-aware	Lonely	Neighbor-aware
Screen on	-1.46E-4	-1.15E-4	-1.24E-4
Network Send	-1.41E-5	-9.1E-6	-9.81E-6
Network Recv	-1.25E-5	-9.11E-6	-9.85E-6
App Service	-1.4E-5	-1.08E-5	-1.02E-5
App foreground	-6.69E-6	-1.1E-5	-1.03E-5

TABLE IV
AVERAGE RANGE COEFFICIENTS

Events	Family-aware	Lonely	Neighbor-aware
Screen on	9.5E5	9.5E5	9.5E5
Network Send	2.2E1	8.45	8.34
Network Recv	1.36E1	4.94	4.9
App Service	6.32E4	6.58E4	7.61E4
App foreground	6.14E4	6.64E4	7.38E4

TABLE V
AVERAGE OFFSET COEFFICIENTS

Events	Family-aware	Lonely	Neighbor-aware
Screen on	0.0	0.0	0.0
Network Send	-1.89E4	-7.94E4	-8.5E4
Network Recv	-1.88E4	-7.94E4	-8.52E4
App Service	-2.06E4	-8.21E4	-8.35E4
App foreground	-2.02E4	-8.3E4	-8.47E4

VI. RESULTS

In the experiments, we tested the different self-organized tweaking strategies (lonely, family-aware and neighbor-aware) on the same dataset, which is collected from a real user by Device Analyzer during a certain time period. This data contains 26 sampling time points, consisting in 8635 collected events including the 26 battery level measurement labels at each time points. We treat the battery measurements as the real smartphone battery level values to evaluate the performance of the MAS energy model. The Figure(3) shows that the particle trajectory shaped by three different tweaking strategies that fits the smartphone battery state of charge (SoC) curve accurately.

To estimate the modeling accuracy at each battery label, we calculate the error between the battery computed by the MAS model and real battery levels. The estimation error metrics is:

$$Error = \frac{B_e - B_m}{B_m} \tag{9}$$

WhereB_e is the estimated battery level, B_m is the measured battery level.

The result in Table(II) shows that the particle trajectory can fit the battery level curve precisely at each sampling time point. It demonstrates that our MAS model has the ability to estimate the state of discharge of the smartphone. Among the three smart-forces strategies, the neighbor-aware smart-forces strategies has the best battery level estimation performance, which suggests that the smartphone energy drain is more related to events captured at near time zone.

The Table(III), Table(IV) and Table(III), show the average strength coefficients, the average strength coefficients and the average strength coefficients calculated by, respectively, family-aware strategy, lonely strategy and neighbor-aware strategy, respectively.

The Table(VI), Table(VII) and Table(VIII), show the average strength deviation, show the average strength deviation and the average strength deviation calculated by, respectively, family-aware strategy, lonely strategy and neighbor-aware strategy. From the results, it can be seen that the average strength coefficients are all negatives, which demonstrates the energy drain effect.

As compared to the results obtained in [15] with manual tuning and with Bezier curves, the results obtained with the tweaking strategies proposed in this paper are more accurate and are a good starting point for further experiments on real datasets.

VII. CONCLUSIONS AND FUTURE WORK

In this paper, we used smartphone user's behavioral data collected by Device Analyzer to construct a smartphone energy consumption model using a multi-agent system paradigm. To that way, a set of agents has been designed for both transferring the data into agent world and to process them using a set of interactions deforming a curve made of particle agents. We also proposed three different strategies for tuning the internal coefficients of the agents. To validate our model, we tested the energy model on real user data. Finally, the result shows that the MAS energy model is capable of modeling energy drain

TABLE VI
STRENGTH STANDARD DEVIATION

Events	Family-aware	Lonely	Neighbor-aware
Screen on	0.0	0.0	0.0
Network Send	2.72E-9	7.47E-6	6.47E-6
Network Recv	3.92E-10	7.46E-6	6.1EE-6
App Service	5.45E-10	9.09E-6	6.27E-6
App foreground	16.55E-7	9.32E-6	6.28E-6

TABLE VII
RANGE STANDARD DEVIATION

Events	Family-aware	Lonely	Neighbor-aware
Screen on	0.0	0.0	0.0
Network Send	1.72	8.03	3.01
Network Recv	8.64E-1	4.68	1.77
App Service	4.21E1	2.76E4	2.71E4
App foreground	3.79E1	2.95E4	2.88E4

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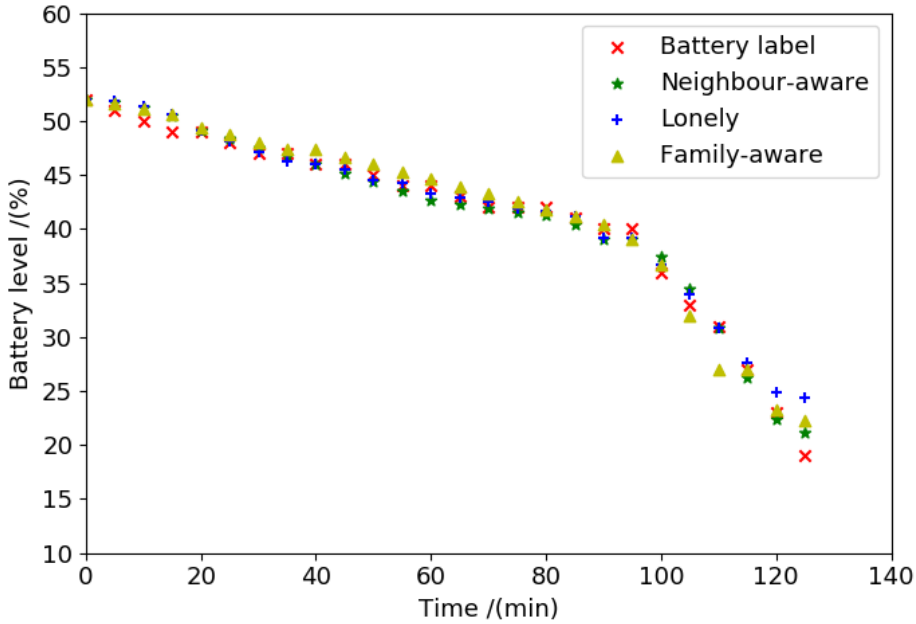

Fig. 3. Estimated Curves

TABLE VIII
OFFSET STANDARD DEVIATION

Events	Family-aware	Lonely	Neighbor-aware
Screen on	0.0	0.0	0.0
Network Send	9.61E-1	2.96E4	2.4E4
Network Recv	1.99	2.97E4	2.43E4
App Service	4.18E1	3.06E4	2.26E4
App foreground	6.7	3.35E4	2.49E4

in the smartphone while proving good results as compared to classical mathematical methods.

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